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Paleomagnetism and $^{40}Ar/^{39}Ar$ ages from La Palma in the Canary Islands

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[1] **Abstract:** The structure of the time-averaged geomagnetic field has been known for centuries to be approximately dipolar. Significant departures of the time-averaged field from that of an axial geocentric dipole, however, have been reported for decades. The data on which time-averaged field models are based must be of the highest quality in order to document subtle long-term features. We present here new paleomagnetic data and $^{40}Ar/^{39}Ar$ ages for the island of La Palma, The Canary Islands. Paleomagnetic samples were obtained from 28 lava flows. Of these, 21 met our minimum acceptance criteria for use in time-averaged field models. The $^{40}Ar/^{39}Ar$ age determinations were successfully carried out on samples from eight of the flows. Our isotopic ages and paleomagnetic polarities are consistent with the currently accepted geomagnetic reversal timescales. The reversed data in the updated database are antipodal to the normal data within the uncertainties, and the time-averaged direction is indistinguishable from that expected from a geocentric axial dipole.

Keywords: Time-averaged geomagnetic field;paleosecular variation;Canary Islands;La Palma paleomagnetism;$^{40}Ar/^{39}Ar$ dating.

Index terms: Reference fields (regional, global); paleomagnetic secular variation; time variations — secular and long term; general or miscellaneous.

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1. Introduction

[2] The fact that the geomagnetic field can be described to first order as resulting from a giant internal bar magnet has been known for hundreds of years. Indeed, this assumption of a geocentric axial dipole (GAD) field is the starting point of much paleomagnetic research, yet significant deviations from a GAD field have been noted for decades [e.g., *Wilson*, 1970].

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[3] Historical observations of the geomagnetic field have been painstakingly collected over the last 2 decades and used to constrain models of the geomagnetic field that average over the time period of the last 400 years [e.g., *Bloxham and Jackson*, 1992; *Jackson et al.*, 2000]. *Johnson and Constable* [1998] and *Constable et al.* [2000] have investigated the average structure of the geomagnetic field over timescales of a few thousand years. All of these models deviate markedly from a simple dipolar field; they exhibit long-term features such as lobes of enhanced radial flux positioned at high-latitude, low temporal variability in the Pacific, and regions of depressed flux near the poles. Conditions at the core-mantle boundary such as lateral temperature variations [*Bloxham et al.*, 1989] or electrical conductivity [*Runcorn*, 1992] have been called on to explain these long-term non-GAD features.

[4] Meanwhile, numerical simulations of the geomagnetic field have recently become much more sophisticated [e.g., *Glatzmaier and Roberts*, 1995; *Kuang and Bloxham*, 1997]. The numerical models of *Glatzmaier et al.* [1999], while still some way from representing the "real" geodynamo, do suggest the possibility of a profound influence of the core-mantle boundary on the behavior of the geomagnetic field. Observations of the time-averaged geomagnetic field over timescales longer than the historical records could place important constraints on the long-term influence of the core-mantle boundary.

[5] Time-averaged field (TAF) models that average over the last five million years rely on compilations of paleomagnetic field observations derived from paleosecular variation studies on lava flows [e.g., *Gubbins and Kelly*, 1993; *Kelly and Gubbins*, 1997; *Quidelleur et al.* , 1994; *Johnson and Constable*, 1995, 1996, 1997, 1998; *McElhinny and McFadden*, 1997; *Carlut and Courtillot*, 1998; *Carlut et al.*, 2000]. The first-order questions that TAF models wish to address are as follows: (1) Are the high-latitude flux lobes seen in the historical geomagnetic field persistent over millions of years? (2) What is the geographic dependence of paleosecular variation? and (3) Are there differences between the normal and reverse polarity average field states? Such questions require data from well-distributed sites over the entire globe that represent both polarities and meet a consistent set of minimum standards for data quality.

[6] In order to begin to address these issues, *McElhinny and McFadden* [1997, hereinafter MM97] compiled a new, comprehensive database of paleosecular variation studies from lava flows. They suggested the following minimum criteria for including a particular study in paleosecular variation models: (1) only data from lavas (and some tuffs) or thin dikes are suitable; (2) there can be no suggestion that the sampling region has been subjected to any tectonic effects; (3) the lava sequence should cover at least 10,000 years; (4) at least five sites should be available for any study of one polarity to be viable (5 + 4 in case of dual polarity); (5) a minimum of $N = 2$ samples per site should have been studied; (6) stability of the magnetization must have been tested by some demagnetization method; (7) the VGP latitude for each site must be $\geq 45^\circ$ (north or south); and (8) the radius of the circle of 95% confidence (α_{95}) for each site must be $\leq 20^\circ$. On the basis of an analysis of the existing database, MM97 concluded that most of the data in the database do not meet even these minimum standards and that the bulk of the studies should be redone. Paleomagnetists have begun the process of updating the paleomagnetic time-averaged field database [see, e.g., *Johnson et al.*, 1998; *Carlut et al.*, 2000]. We present here new results from the Plio-Pleistocene rocks of the Island of La Palma in the Canary Islands.

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2. Previous Work in the Canaries

[7] MM97 describe a database of paleosecular variation studies that have been done on Plio-Pleistocene (0–5 Ma) lava flows. This database can be downloaded from ftp.ngdc.noaa.gov and contains data from the Canary Islands, an archipelago off the west coast of Africa (Figure 1a). The first paleomagnetic investigations in the Canary Islands were made by Watkins and colleagues in a series of papers [e.g., *Watkins et al.*, 1966; *Abdel-Monem et al.*, 1971, 1972] that culminated in a paper summarizing the Canarian data set [*Watkins*, 1973, hereinafter W73]. These studies have been supplemented recently [e.g., *Carracedo and Soler*, 1995, hereinafter CS95; *Guillou et al.*, 1996, hereinafter G96; *Quidelleur and Valet*, 1996 , hereinafter QV96] and have been included in the most recent release of the database (August 1998 as of this writing).

[8] There are 128 paleomagnetic directions from the Canarian paleomagnetic database that meet the minimum criteria of MM97. These are shown in Figure 2a. To first order, the questions we wish to ask of the data are as follows: (1) Can the reversed and normal data be considered antipodal? (2) Can either data set be distinguished from the direction expected from a geocentric axial dipole? and (3) How robust are the conclusions and what effect do stricter selection criteria have on the outcome?

[9] In order to address these issues, we must employ statistical techniques. The most straightforward way of addressing the problem of antipodality (also known as the "reversals test") is using the methods developed by *Fisher* [1953]. We list the Fisher parameters in Table 1 (see also Figure 1). The confidence ellipse of the antipodes of the reversed data includes that of the normal data, but neither confidence ellipse includes the mean direction of the other data set; the reversals test is equivocal. This sort of "gray zone" case has been addressed by, for example, *Watson* [1956, 1983, 1984], *McFadden and Lowes* [1981], *McFadden and McElhinny* [1990] and *Tauxe* [1998]. A particularly useful parameter, *Vw*, was proposed by *Watson* [1983]. It can be defined as follows.

[10] For the *i*th data set with N_i unit vectors whose direction cosines are x, y, and z, we define

$$\overline{X}_i = \sum_j^{N_i} x_j, \qquad \overline{Y}_i = \sum_j^{N_i} y_j, \qquad \overline{Z}_i = \sum_j^{N_i} z_j.$$

The resultant vector R_i of the data set is, of course,

$$R_i = (\overline{X}^2 + \overline{Y}^2 + \overline{Z}^2)^{\frac{1}{2}}.$$

When combining M data sets with different precision parameters k_i (estimated by $k_i = (N_i - 1)(N_i - R_i)$, we take the weighted means

$$\hat{X} = \sum_j^M k_j \overline{X}_j, \qquad \hat{Y} = \sum_j^M k_j \overline{Y}_j, \qquad \hat{Z} = \sum_j^M k_j \overline{Z}_j,$$

and define the weighted overall resultant vector Rw as

$$Rw = (\hat{X}^2 + \hat{Y}^2 + \hat{Z}^2)^{\frac{1}{2}}$$

and the weighted sum Sw as

$$Sw = \sum_j^M k_j R_j.$$

[11] Finally, Watson's Vw is defined as

$$Vw = 2(Sw - Rw).$$

[12] Vw was posed as a test statistic that increases with increasing difference between the mean directions of the two data sets. Thus the null hypothesis that two data sets have a common mean direction can be rejected if Vw exceeds some critical value.

[13] *McFadden and McElhinny* [1990] proposed that the critical value could be deter-

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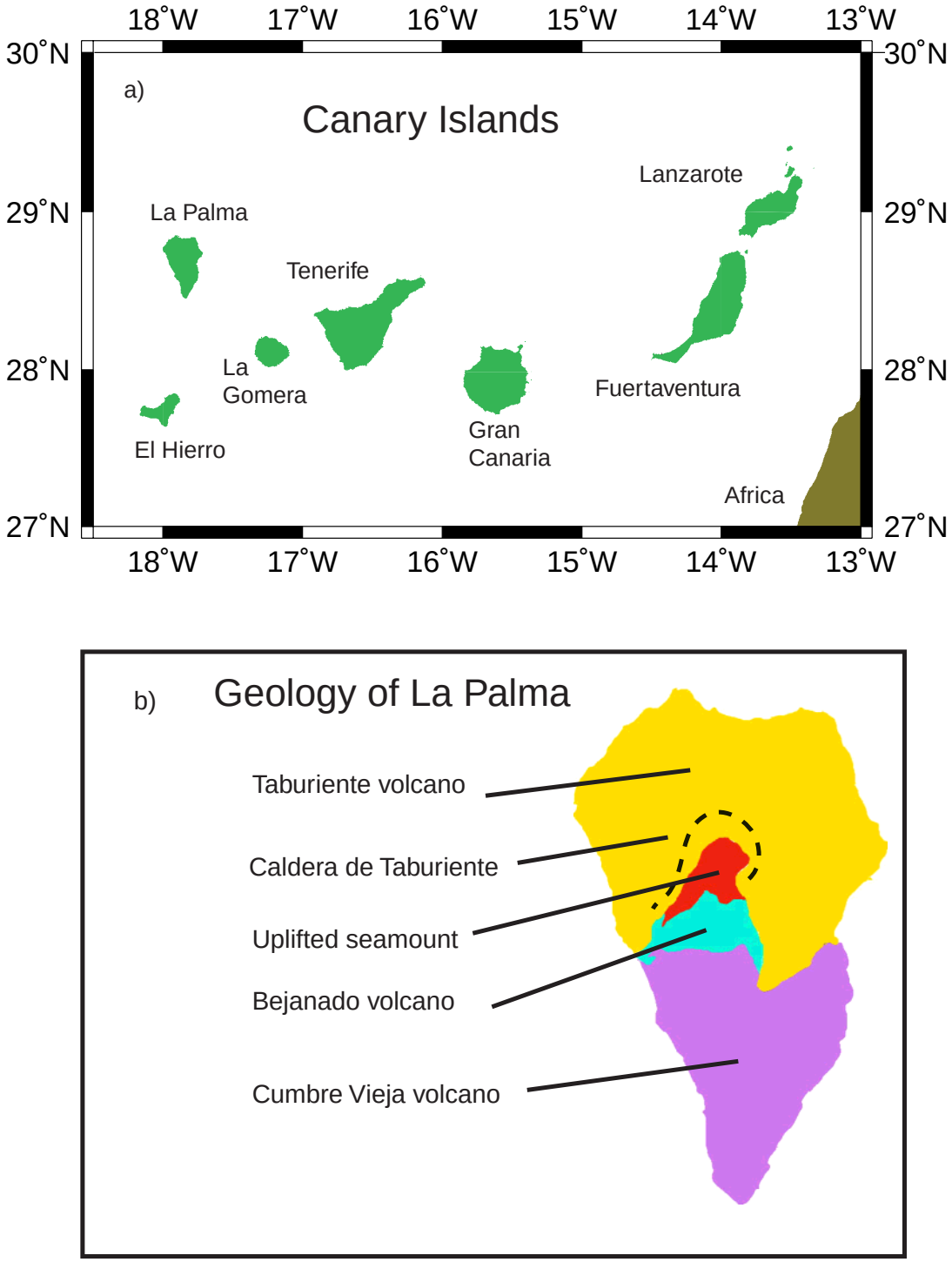


Figure 1. (a) Map showing the Canary Islands Archipelago off the west coast of Africa. La Palma is the northwesternmost island. (b) Sketch of the main geological units of La Palma (redrawn from *Carracedo et al.* [1999].

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Table 1. Summary Statistics for Published Canarian Paleomagnetic Data Meeting MM97 Criteria[a]

Polarity	$\overline{D}$	$\overline{I}$	α_{95}	N	k	Fisherian?
Normal	6.8	42.3	3.1	90	24	yes
Reversed	177.7	−38.7	6.8	38	13	no
Combined	4.1	41.3	3.0	128	19	no
IGRF	351.2	39.1				
GAD	0	47				

[a]IGRF, International Geomagnetic Reference Field; GAD, geocentric axial dipole.

mined through simulation. Thus we calculate *Vw* for the normal and the antipodes of the reversed data shown in Figure 2a as 6.65. In order to determine the critical value for *Vw*, we draw two Fisher distributed data sets with dispersions of k_1 and k_2 but having a common direction. We use the method described by *Fisher et al.* [1987] to generate Fisher distributions; see also *Tauxe* [1998] . We calculate *Vw* for these simulated data sets, repeating the procedure 1000 times. The resulting *Vw* values are plotted in a histogram in Figure 2c with the bounds containing the lowermost 95% shown as a dashed line. The value for the Canary database (6.65), shown as a heavy vertical line, is larger than the critical value of 6.19. This procedure therefore supports the suggestion that the two data sets are not antipodal at the 95% level of confidence.

[14] The Fisher-Watson-McFadden genre of tests relies on the directional data sets being *Fisher* [1953] distributed. Unfortunately, the reversed data shown in Figure 2a are not. A Fisher distribution demands that the declination data be uniformly disposed about the principal direction and that the coinclination data be exponentially disposed away from it. As described by *Fisher et al.* [1987], data can be plotted against the value expected for a particular distribution in a so-called quantile-quantile plot. If the distribution is appropriate, the data plot is a straight line. Significant deviations from linearity can be quantified using tailored Kolmogorov-Smirnov parameters. For the purposes of testing whether a given data set is Fisherian, the appropriate statistics are M_u and M_e, whose critical values are 1.207 and 1.094, respectively. The data in Figure 2a are shown as a quantile-quantile plot in Figure 3. The reversed mode fails as shown in Figure 3d (see *Fisher et al.* [1987] and *Tauxe* [1998] for more details). Therefore tests based on a Fisher assumption may not be applicable to the question at hand.

[15] It was for data such as these that a bootstrap reversals test was developed [*Tauxe et al.*, 1991; *Tauxe*, 1998]. The bootstrap is in many ways similar to the simulations just described, but instead of drawing Fisher distributions with characteristics similar to those of the original data set, we simply draw so-called bootstrapped data sets by randomly selecting N directions from the original data; some directions may be used more than once, while others are not used at all. Instead of assuming an inappropriate distribution, we assume that the data set is large enough to represent the underlying distribution adequately.

[16] The Fisher mean [*Fisher*, 1953] is calculated for the normal and reversed modes of each bootstrapped data set. The direction cosines of the "bootstrapped" means of each parasample are plotted in Figure 4, taking the antipodes of the reversed means. The reversed data can be compared with the normal data and both can be compared with the GAD direction at La Palma (shown as the solid lines in

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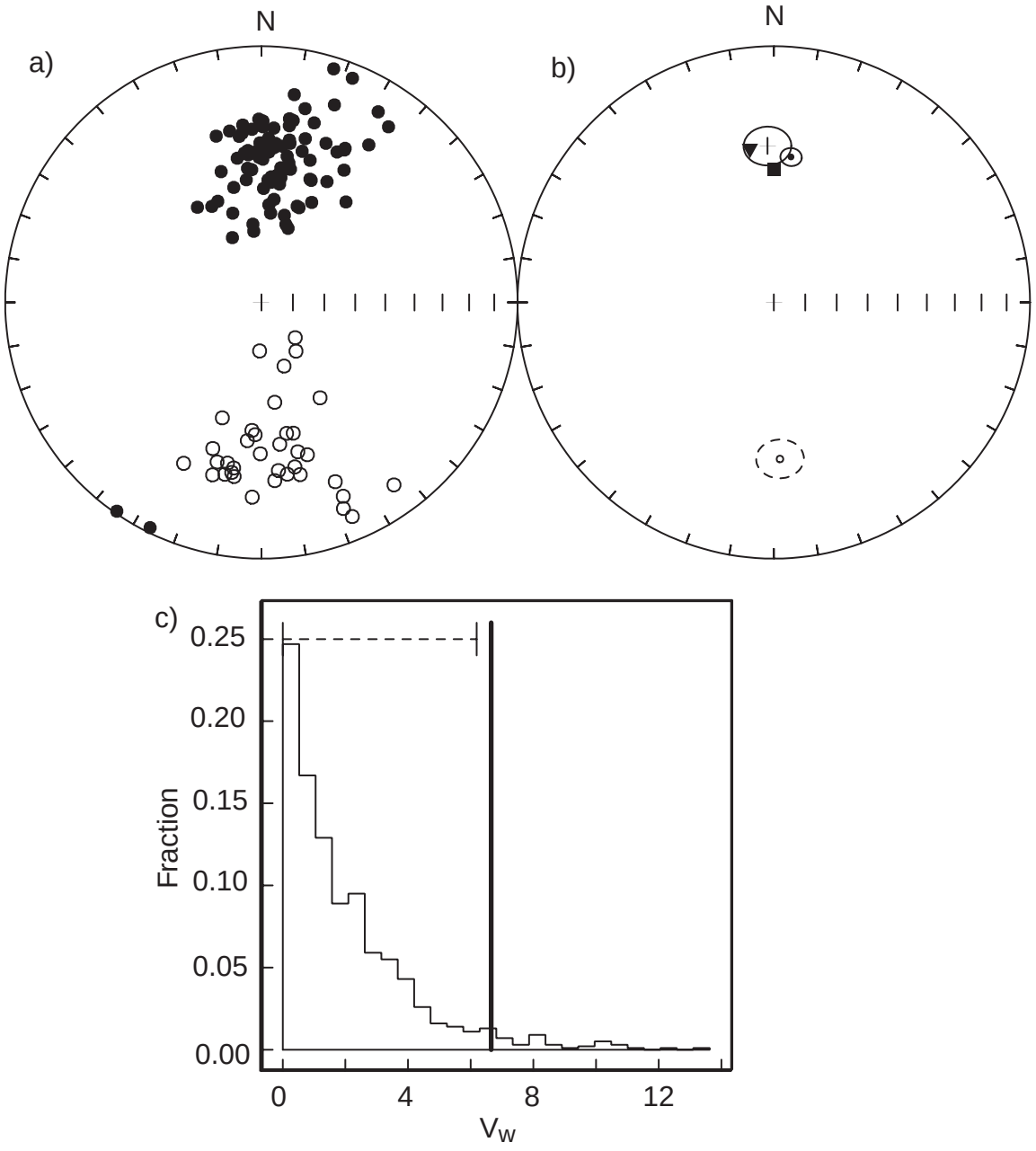


Figure 2. (a) Equal area projections of declinations and inclinations of Pleistocene data from the Canary Islands meeting the minimum criteria of *McElhinny and McFadden* [1997]. Solid (open) symbols are lower (upper) hemisphere projections. (b) Fisher means from the data shown in Figure 2a. The solid (open) circle is the mean normal (reverse) direction, the plus sign is the antipode of the mean reversed direction, the square is the expected geocentric axial dipole (GAD) direction and the triangle is the International Geomagnetic Reference Field (IGRF) direction in La Palma in 1993 (see Table 1). Ellipses are the *Fisher* [1953] α_{95} values. (c) Distribution of V_w for simulated Fisher distributions. The dashed line includes the smallest 95% of the V_ws calculated for the simulations. The heavy vertical line is the V_w calculated for the normal and reversed (antipodes) data sets. According to this test, the two data sets are not antipodal.

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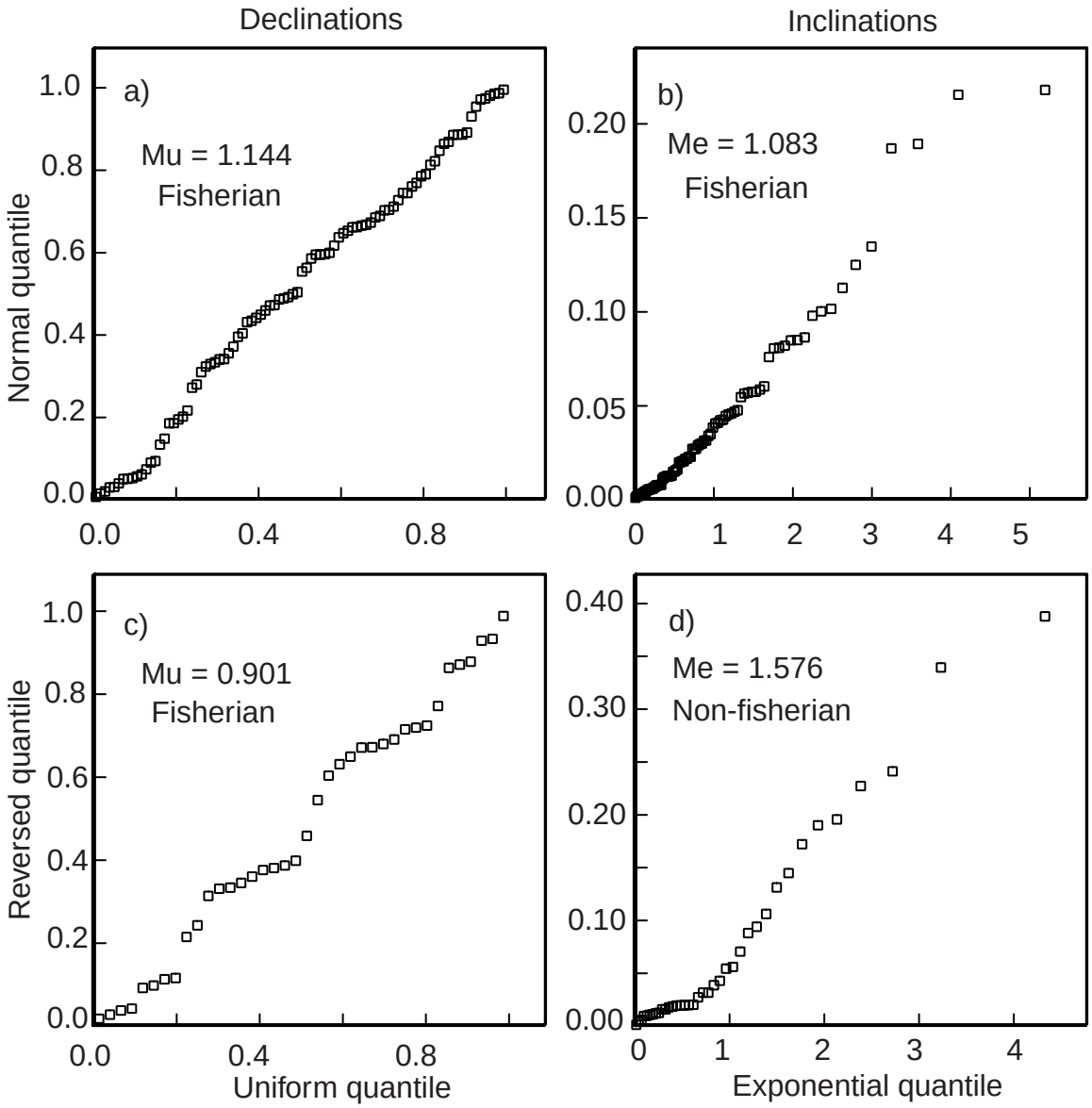
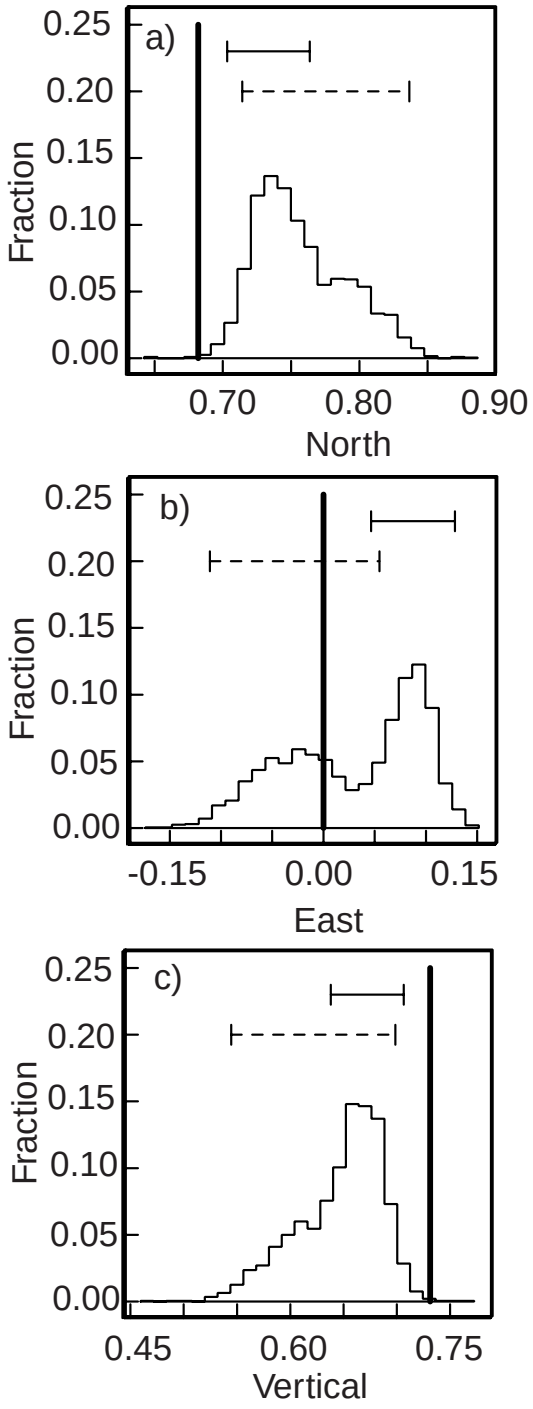


Figure 3. Quantile–quantile plots for the MM97 selection of Canarian paleomagnetic data from Figure 2a: (a, b) normal data; (c, d) reversed data.

Figures 4d–4f). The bounds containing 95% of the bootstrapped components are drawn above histograms of the bootstrapped components (bounds for the reversed data are indicated by the dashed line). These bounds constitute bootstrap confidence intervals. The normal and reversed modes appear to be antipodal (the bootstrap confidence intervals do not overlap for the north Cartesian component (see Figure 4). Please note that the principal drawback of bootstrap methods is a tendency to underestimate uncertainty if the data set or the number

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of parasamples drawn is too small. Thus our result of a positive reversals test is a conservative one. More data would only tend to make the confidence intervals bigger. Thus our bootstrap result, which is in conflict with the parametric method of *Watson* [1983] as enhanced by *McFadden and McElhinny* [1990], underscores the difficulty of applying Fisherian tests to non-Fisherian data.

[17] Also shown in Figure 4 is a comparison of the bootstrapped confidence intervals and the expected components from a GAD field. Neither mode appears to be consistent with the expected GAD direction, whose components are indicated by the heavy vertical lines in the component histograms.

[18] We turn now to a discussion of the published data from the Canaries and consider whether the conclusions based on them may be biased in some way. For example, there may be overprinting, bias from structural rotations, or bias from sampling strategy that may affect the data distributions in the database.

[19] The W73 data constitute 234 site means for the Plio-Pleistocene. More than half of these were based on $N = 2$ samples. While all specimens were cleaned at several alternating field (AF) steps, site means were based on selecting the steps that yielded the tightest clustering of data. Many of the sites fail the α_{95} criterion of

Figure 4. Histograms of the direction cosines of the mean vectors of simulated data sets bootstrapped from Figure 2a (antipodes of the reversed means are shown): (a) north components, (b) east components, and (c) vertical components. The intervals indicated above the histogram are the bounds containing 95% of the normal (solid) and reversed (dashed) components. The heavy vertical line is the north component of the expected GAD direction at La Palma. According to this test, the reversed and normal data are antipodal, but both are distinct from the expected GAD direction.

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MM97 and only 68 were selected for the MM97 database.

[20] The CS95 data set comprises 13 sites. At least one specimen per site was subjected to stepwise AF and thermal demagnetization, and the rest of the samples were cleaned at a single "optimal" step. All site means were based on at least $N = 4$ samples, and the largest α_{95} reported was 6.4°.

[21] The G96 data set includes 26 lava flows. These were apparently treated in a fashion similar to the CS95 samples, although the details of the demagnetization procedure are not clear and no examples are shown. Five of the site means were based on only $N = 3$ samples. According to the authors, the "El Golfo Section" represents the head wall of a major volcanic collapse structure that created the El Golfo embayment. This contention is supported by the discovery of large masses of landslide debris on the seafloor offshore from the El Golfo embayment [*Masson and Watts*, 1995]. The El Golfo lavas are distinguished from those in the northeastern rift zone by having opposing dips of the lava flows. In the El Golfo escarpment the general structure is periclinal, pointing to a volcanic edifice centered in the present-day El Golfo embayment. As described by *Guillou et al.* [1996], renewed tectonic activity began in the younger part of the Brunhes chron, further disrupting the El Golfo sequence. The activity of a triple rift system has continued until recently and has covered the island with recent vents and lavas that totally or partially fill the tectonic scars. Overgrowth instability resulted in the successive gravitational collapse of the El Golfo section. Finally, the Tiñor Section of *Guillou et al.* [1996] is on the "San Andres fault system," and some of the paleomagnetic data are from "steeply dipping lavas." On this basis the El Hierro data set as a whole might be suspected of significant tectonic disturbance, and the El Golfo sequence, in particular, should not be used for time averaged field modeling.

[22] Several data sets have been published recently that have not yet been included in the database. These include *Széréméta et al.* [1999] and *Valet et al.* [1999]. We will now consider whether any of these are suitable for inclusion in TAF models.

[23] *Széréméta et al.* [1999] reinvestigated the El Golfo Section of El Hierro for the purposes of constraining the time-averaged field. The S99 data are rotated clockwise and are shallower than the GAD direction. The authors neglect possible tilting and rotate the data back along a vertical axis. We believe that the evident structural complications discussed by *Guillou et al.* [1996], bolstered by the seafloor mapping data of *Masson and Watts* [1995], render the data from El Hierro, and, in particular, the El Golfo Section, suspect; they probably should not be included in our database for time-averaged field modeling, as they fail criterion 2 of MM97.

[24] *Quidelleur and Valet* [1996] and *Valet et al.* [1999] studied sections known to contain transitional flows. The resulting paleomagnetic data, which were not collected for the purposes of constraining TAF models, may bias time-averaged field models in two ways. First, the data set includes directions that are intermediate between normal and reversed and should be excluded from the time averaged field data base. Second, these data come from continuously sampled sequences of lava flows (and sills) whose age relations are poorly constrained. It is difficult to avoid overemphasis of particular geomagnetic field states and to ensure that each data point represents a distinct estimate of the geomagnetic field, and efforts at combining data from sequential lava flows are fraught with

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difficulty [e.g., *Love*, 2000]. In order to avoid such difficulties, *Johnson and Constable* [1998, p. 95] suggested that data for paleosecular variation studies should have "sufficient information in the reference to establish temporal independence of the flows." Hence these data too should not be included in TAF models.

[25] Thus data from the Canary Islands are still needed for TAF models. We present here results obtained from lava flows exposed on the island of La Palma.

3. Sampling

[26] The geology of La Palma was summarized by *Carracedo et al.* [1999]. La Palma, on the western edge of the archipelago and one of the youngest islands (see Figure 1a), was constructed during the Plio-Pleistocene. The geology of La Palma (see Figure 1b) is the work of four main volcanic centers: the seamount, the Taburiente volcano, the Bejando volcano, and the Cumbre Vieja volcano. The oldest rocks, found in the Caldera de Taburiente, are the gabbros, dikes, submarine volcanics, and sedimentary units of an uplifted and exhumed Pliocene seamount [*Staudigel and Schminke*, 1984].

[27] The subaerial lavas (and some thin dikes and sills) of the Taburiente volcano range in age from perhaps as old as 2 Ma [*Ancochea et al.*, 1994] to some 566 ka [*Guillou et al.*, 1998]. The Taburiente volcano suffered a major collapse in an enormous landslide event that initiated the excavation of the Caldera de Taburiente ~560 kyr ago [*Carracedo et al.*, 1999]. Postcollapse volcanism generated the Bejando volcano between ~560 and 500 kyr ago. Over the last ~125 kyr, volcanism has been associated with the Cumbre Vieja volcano with activity as recent as 1971.

[28] For the purposes of providing data constraining the time-averaged geomagnetic field, we found the northern exposures of the Taburiente volcano to be an irresistible target. The rock magnetic properties have been well characterized by *Quidelleur and Valet* [1996], *Guillou et al.* [1998], and *Valet et al.* [1999]. While not always ideal for paleointensity studies (*Quidelleur and Valet* [1996] and L. Tauxe (data from all attempted Thellier-Thellier experiments for the La Palma samples (lpthellier.pdf), available at http://www.g-cubed.org), but see *Valet et al.* [1999]), these lavas provide excellent material for studying the directional properties of the geomagnetic field. Furthermore, there are readily accessible exposures of lava flows along the road that traverses the northern portion of La Palma.

[29] We were particularly interested in documenting the time-averaged reversed field in the Canaries, which is less well represented in the published database. Probable polarities were determined using a Bartington flux gate magnetometer. We also used field relations such as the presence of paleosols, which indicate significant geological time separation, to ensure that each lava flow that we sampled constituted a unique measurement of the geomagnetic field.

[30] Twenty-nine flows or thin dikes were sampled for this study in December 1993. Flow locations are listed in Table 2 and are shown in Figure 5. From 5 to 12 separately oriented samples were obtained from each lava flow using a gasoline-powered drill. Some sites have regrettably fewer samples than the more recent field studies sampled by us [e.g., *Johnson et al.*, 1998], in which we routinely take at least 10 samples.

[31] Our general strategy is to take samples from as wide a spacing as possible in the outcrop. Our samples therefore span from several meters to some 10 m of outcrop extent.

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Table 2. Summary of New Paleomagnetic and Age Data From La Palma

Site	Latitude, N / Longitude, W	Age	*N*	*D*	*I*	α_{95}	Polarity
LP1001	28°42.31′/17°57.08′	Cln	5	356.6	38.5	3.1	N
LP1002	28°42.31′/17°57.08′	Cln	4	352.7	28	2.9	N
LP1003	28°42.31′/17°57.08′	Cln	5	329.3	48.1	2.9	N
LP1004	28°46.49′/17°57.82′	Cln	5	216.1	62.3	45.1	X
LP1005	28°49.90′/17°56.42′		5	1.2	15.5	11.4	X
LP1006	28°49.93′/17°56.55′	Cln	5	358.5	39.9	3.3	N
LP1007	28°49.93′/17°56.55′		6	9	15.5	14.9	X
LP1008	28°39.60′/17°56.32′	Clr.lr	6	169.2	−28	5.1	R
LP1009	28°39.60′/17°56.32′	0.9714 ± 0.038	5	163.3	−45.6	3.5	R
LP1010	28°39.22′/17°56.84′	0.87 ± 0.02	5	200.9	−27.9	4.1	R
LP1011	28°47.11′/17°57.94′		5	305.5	37.9	12.4	X
LP1012	28°48.70′/17°57.70′		8	201.9	−63	30	X
		a) 1.156 ± 0.055					
		b) 1.100 ± 0.025					
LP1013	28°48.98′/17°57.67′	Cln	4	5.3	34.9	3.2	N
		a) 0.615 ± 0.014					
		b) 0.604 ± 0.022					
LP1014	28°48.85′/17°57.55′	Clr.lr	5	189.2	−43.4	2.7	R
		a) 0.858 ± 0.035					
		b) 0.848 ± 0.035					
LP1015	28°48.85′/17°57.55′	Clr.lr	6	191.4	−32.5	3.4	R
LP1016	28°48.85′/17°57.55′	Cln	4	353.2	59.8	4.7	N
LP1017	28°47.04′/17°51.48′		7	36.8	42	24.3	X
LP1018	28°47.7′/17°51.35′	C2n[a]	5	343.4	43.9	2.8	N
LP1019	28°48.55′/17°49.85′		5	176	−38.7	2.5	R
LP1020	28°50.18′/17°49.85′		6	218.2	−2.1	9.6	X
LP1021	28°50.22′/17°52.80′		5	161.3	−42	4.1	R
LP1023	28°47.95′/17°50.05′		5	170.2	−38.4	2.9	R
LP1024	28°47.95′/17°50.05′		5	5.2	58.9	2.9	N
LP1025	28°48.55′/17°49.85′		7	195.2	−22	4.7	R
LP1026	28°47.91′/17°46.55′	0.86 ± 0.02	5	163.1	−43.1	2.7	R
LP1027	28°47.91′/17°46.99′		6	43.4	11.1	13.1	X
LP1028	28°47.21′/17°46.09′		5	199.3	−37.7	1.8	R
LP1029	28°43.50′/17°45.09′		6	184.5	−60	3.9	R
		a) 1.210 ± 0.466					
		b) 1.150 ± 0.027					
LP1030	28°43.50′/17°45.09′	1.034 ± 0.03	4	18.4	60.9	3.9	N

[a]Same location as P.15.C(1.79 ± 0.06) of *Ancochea et al.* [1994].

This is done to ensure that the average direction obtained is not affected by local magnetic field anomalies, distortions, or reheating events.

[32] All cores were oriented using a magnetic compass. Magnetic compass azimuths were adjusted for the magnetic declination at La Palma (see International Geomagnetic Reference Field (IGRF) value in Table 2). Sun compass measurements were also obtained for about one third of the samples. The difference between the sun compass and the (declination-adjusted) magnetic compass measurements was negligible. We therefore used the magnetic azimuths (adjusted for the IGRF (1993) declination in La Palma) for orienting all samples.

4. The $^{40}Ar/^{39}Ar$ Analysis

[33] Samples from eight sites were selected for $^{40}Ar/^{39}Ar$ analysis in the Vrije Universiteit laboratory. Thin sections were examined for

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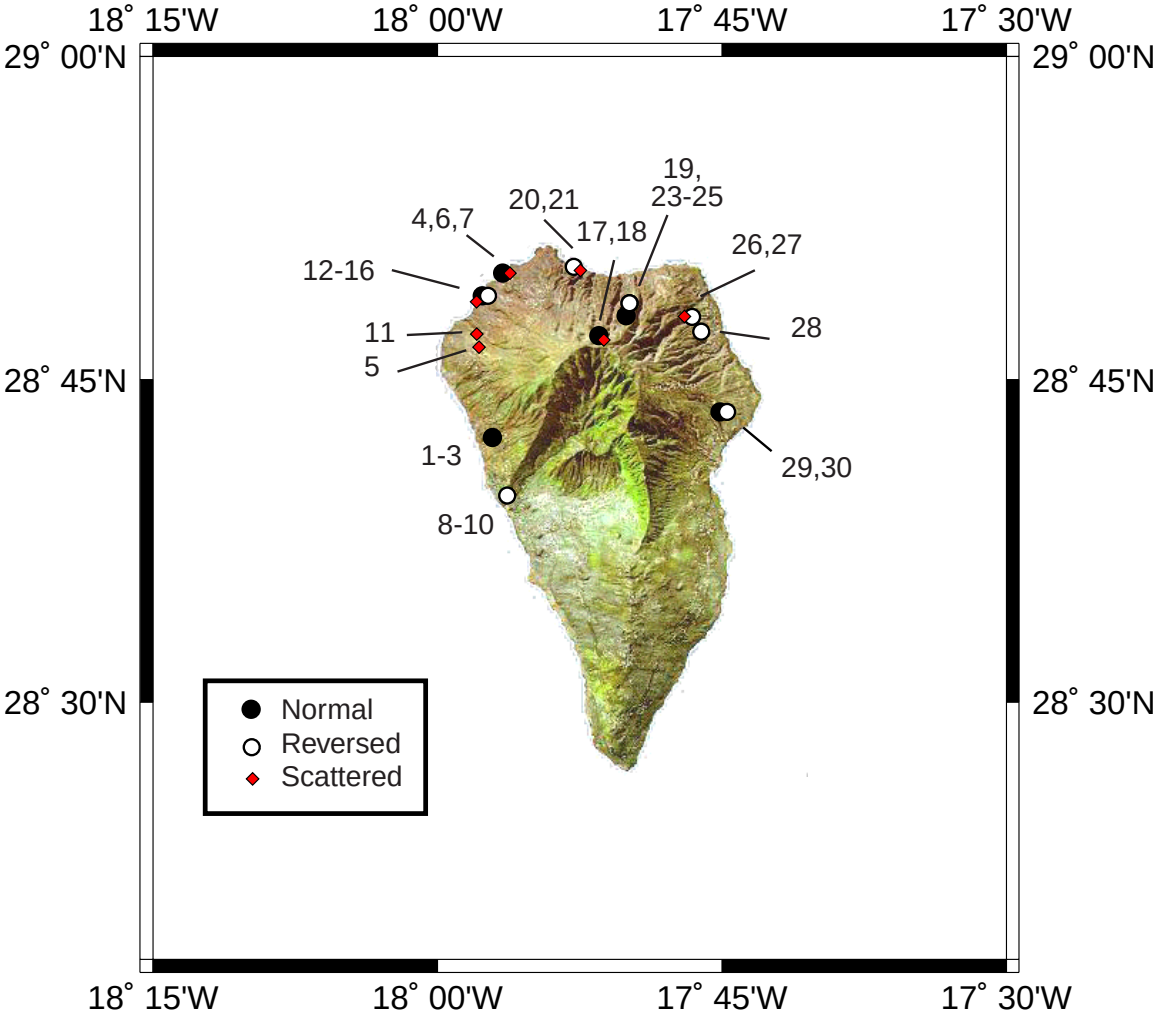


Figure 5. Map of site locations from Table 2. Site numbers refer to the LP1000 series. Black symbols are normal sites, white symbols are reversed sites, and red diamonds are sites not meeting the minimum acceptance criteria.

the presence of glass, hydrothermal alteration, and mafic phenocrysts. The approach followed in this study is similar to that of *Johnson et al.* [1998]. Four groundmass samples underwent incremental heating analysis while all eight were prepared from the 25-mm drill cores by drilling minicores of 5-mm diameter. From these minicores we prepared ~1-mm-thick slices that were stacked in quartz vials for irradiation purposes. The groundmass separates were loaded in aluminum foil packages for irradiation. These foil packages were then stacked in quartz vials. Laboratory standard sanidine DRA-2 (age 26.36 Ma, with respect to GA 1550 standard biotite at 98.79 Ma [see *Wijbrans et al.*, 1995; *Renne et al.*, 1998]) was used to monitor the neutron flux. Standards were packed in Cu foil and inserted between every five unknown minicore slices or three foil packages (groundmass separates).

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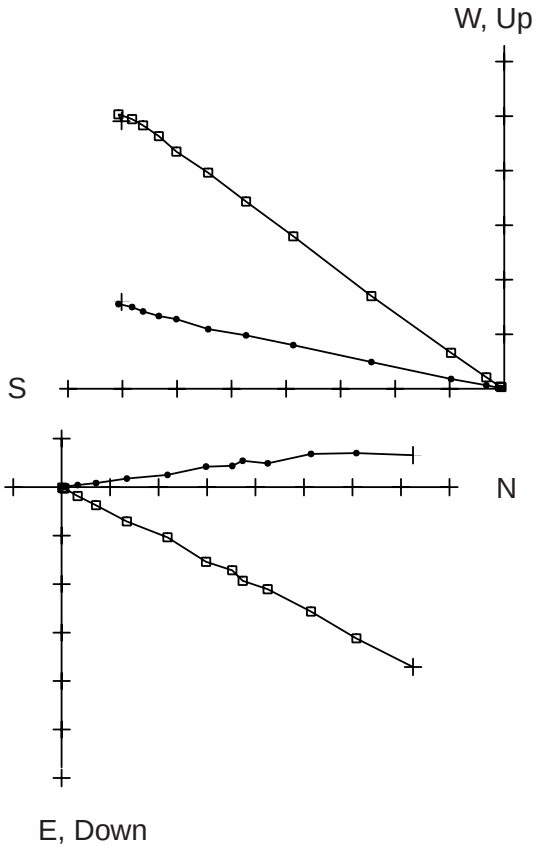


Figure 6. Orthogonal projections of representative well-behaved specimens during thermal demagnetization. Horizontal (vertical) projections are solid (open) symbols.

[34] The samples were irradiated for 1 hour in the Oregon State University TRIGA reactor Cadmium-lined, in-core irradiation tube (CLICIT) facility. Replication of the standards was generally better than 0.3% (one standard deviation). By fitting a second-order polynomial through the standards data against position in the sample tube, one can find the effective J value appropriate for the position of the unknown with an equivalent uncertainty.

[35] For the incremental heating experiments the laser power was increased from 1 to 8 W typically over 7 to 8 steps, with the laser defocused to a spot of ~3-mm diameter. The 30-mg groundmass separates were spread out thinly on the bottom of 13-mm diameter copper pans, and homogeneous heating was obtained by slowly circling the laser beam over the bottom of the pan. Corrections for interfering reactions from K and Ca were those reported by *Wijbrans et al.* [1995]. Radioactive decay of ^{37}Ar and ^{39}Ar and mass spectrometer discrimination (1.003±0.1% per mass unit (pmu)) determined from repeated analysis of small aliquots of atmospheric argon were applied. Blank runs were measured before and after each fusion step for the single fusion experiments and every second or third step for the incremental heating experiments. Blank values appropriate for each experiment were interpolated by regression of measured blank intensity against time during the day. Calculation of the plateau ages required that at least three steps, which together compose more than 50% of the total gas release yielded, ages within two standard deviations of the plateau age.

[36] The incremental heating experiments (see appendix tables) demonstrated that reasonable, undisturbed plateaus could be obtained when the first and final steps were omitted from the calculation. This approach was then adopted for the minicore slices. The amounts of material did not allow us to carry out incremental heating experiments on the minicore slices. Instead, these were heated in three steps, of which the first and final steps were enriched in atmospheric argon contaminant and consequently rejected for deriving age information. While incremental heating produces better data in general, we nonetheless feel that the fusion data for the four that did not undergo full incremental treatment are reasonably reliable. We therefore list data for all eight lava flows. Please note that two fusion heating experiments were done for each flow and the replicate

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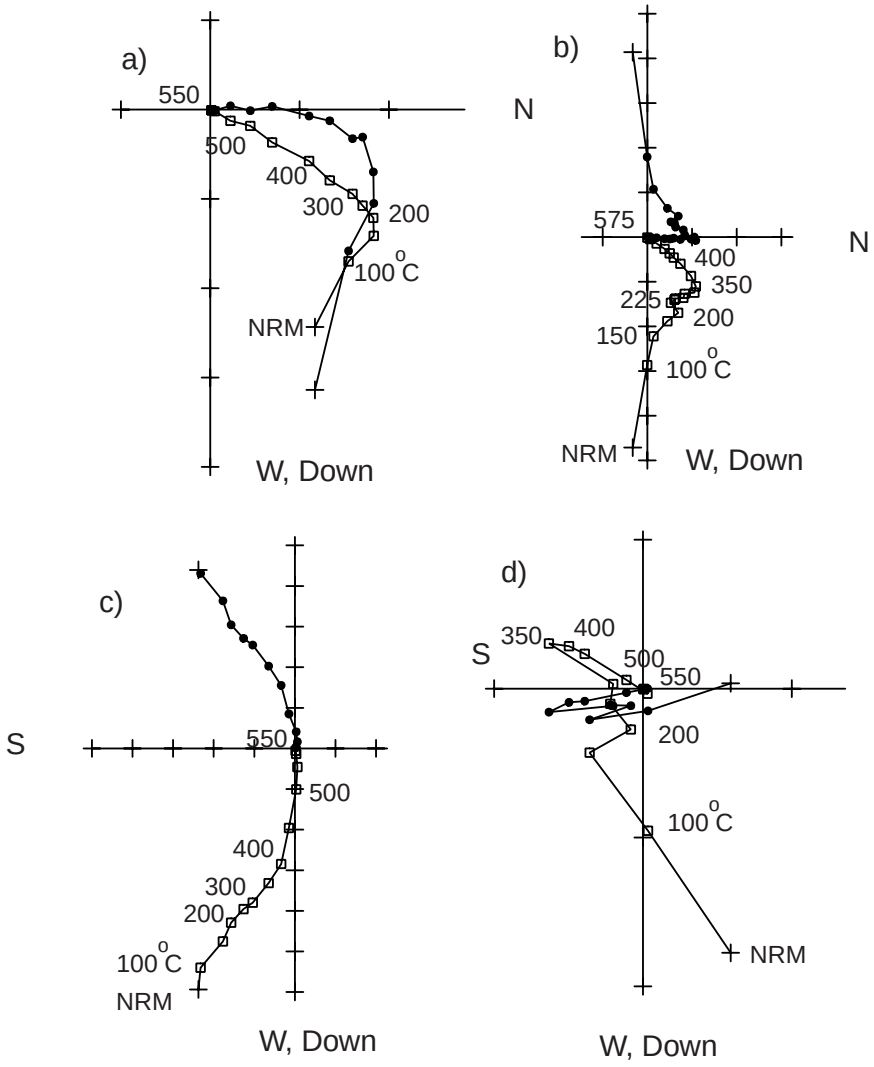


Figure 7. Orthogonal projections of representative ill-behaved specimens during thermal demagnetization. Same conventions as in Figure 6.

estimates agree with each other within the quoted uncertainties.

5. Paleomagnetic Analysis

[37] At least five samples from each of the 29 sites were sliced into 2.5-cm cylinders for paleomagnetic analysis. The natural remanent magnetization (NRM) of one specimen from each sample was measured in the Scripps Paleomagnetic Laboratory using a CTF three-axis cryogenic magnetometer.

[38] After measurement of the NRM a single specimen from at least five samples from each site was subjected to stepwise thermal demagnetization at 50° steps starting at 100°C up to 500°C, followed by 25° steps until maximum unblocking was achieved at 575°C. Demagne-

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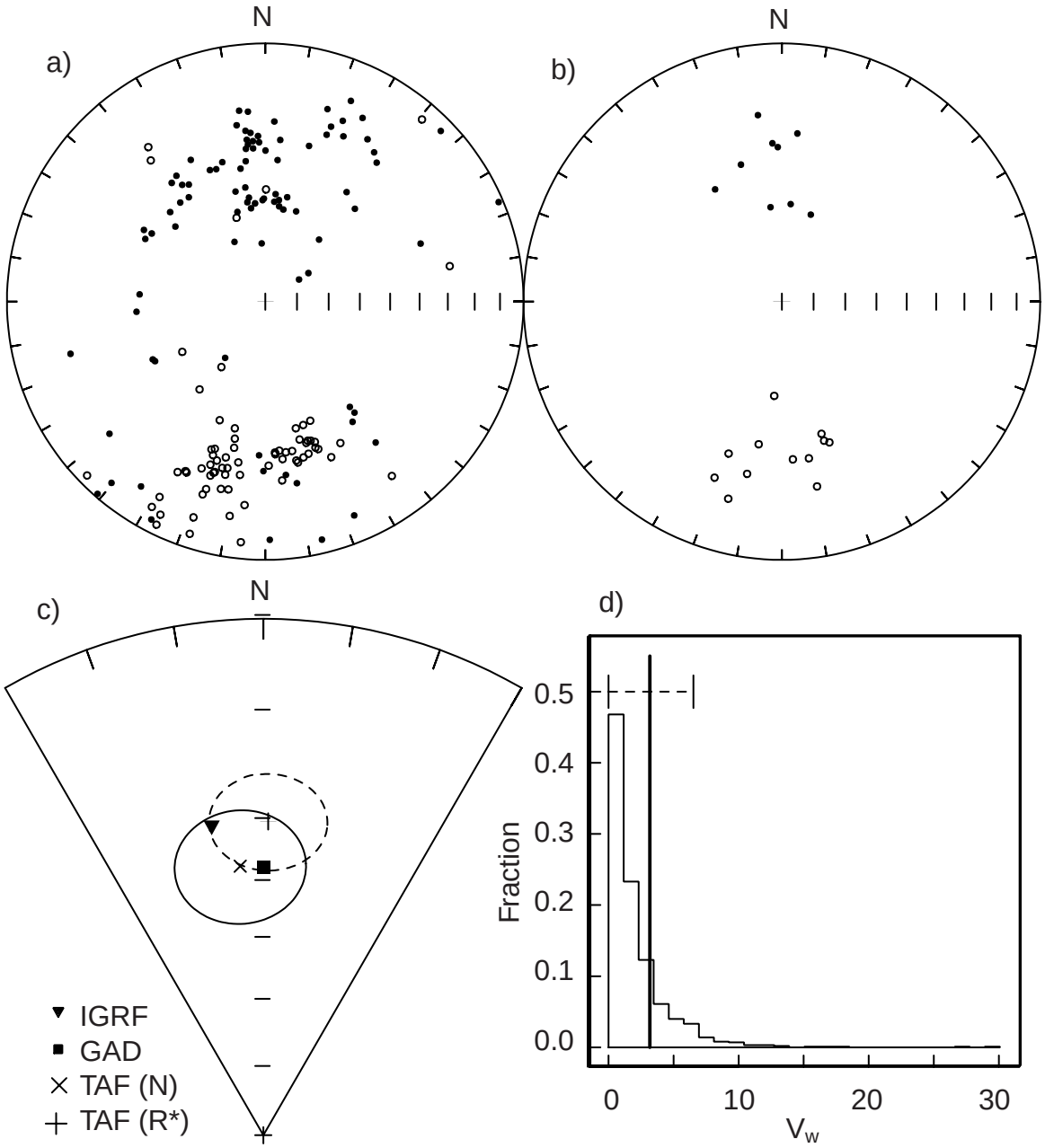


Figure 8. Equal area projections of declinations and inclinations of the La Palma data set (see Table 2). (a) NRMs from this study. (b) Site mean directions from this study with at least four samples meeting minimum criteria and $\alpha_{95} \leq 5$ (rounded to the nearest integer). (c) *Fisher* [1953] mean directions of the normal, and the antipodes of the reversed sites, with 95% confidence ellipses. Also shown are the GAD and IGRF directions for La Palma. (d) Distribution of *Vw* for simulated Fisher distributions. The dashed line includes 95% of the *Vw* values calculated for the simulations. The heavy vertical line is the *Vw* calculated for the normal and reversed (antipodes) data sets. According to this test the two data sets are antipodal.

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Table 3. Summary Statistics for LP Data Set

Polarity	$\overline{D}$	$\overline{I}$	α_{95}	N	k	Fisherian?
Normal	355.1	46.6	9.6	9	30.	yes
Reversed	180.9	−39.1	8.4	12	28.	yes
Combined	358.6	42.4	6.1	21	28	yes

tization data were classified into two categories: well-behaved (see examples in Figure 6) and ill-behaved (see examples in Figure 7). All demagnetization data (magnetometer data from LP samples (lp.dat), Zijderveld diagrams for the data in lp.dat (lpdmag.pdf), and site means from acceptable characteristic directions (lp.dfs)) are available at http://www.g-cubed.org.

[39] Acceptable characteristic remanence directions met the following minimum criteria: (1) At least five demagnetization steps were used to calculate a principal component using the method of *Kirschvink* [1980]. (2) The maximum angle of deviation (MAD) for the data about the principal direction was $\leq 5^\circ$. (3) The demagnetization data defining the characteristic direction must (by definition) trend toward the origin, within uncertainty.

[40] Examples of specimens whose demagnetization data failed to meet these minimum criteria are shown in Figure 7. Some suffer from curved demagnetization trajectories resulting from overlapped blocking temperatures of the characteristic component and some other direction (e.g., Figures 7a and 7c). Other specimens (e.g., Figures 7b and 7d) display more complicated demagnetization diagrams perhaps resulting from thermal remagnetization by overlying lava flows as described by *Valet et al.* [1998]. Both behaviors were exhibited by both polarities.

[41] As ill-behaved specimens are not uncommon, we worry about studies in which specimens are treated by some ''blanket'' or ''optimal'' cleaning technique. The single-step treatment could mask underlying complications. In our experience a large amount of scatter results from inclusion of such data. The criterion that α_{95} values must be less than 20° is not sufficient protection. For this reason, we have adhered to the principal component criteria mentioned above.

[42] Site mean directions were calculated using *Fisher* [1953] statistics for flows with specimens from at least four samples that yielded acceptable principal components. These mean directions were deemed acceptable for the purpose of constraining the time-averaged field if the site means had α_{95} values of ≤ 5 (rounded to the nearest integer). The paleomagnetic results are summarized in Table 2.

[43] NRM directions are plotted in equal area projection in Figure 8a. All acceptable site means directions are plotted as circles in Figure 8b. The scattered nature of the NRM in the La Palma samples is substantially reduced by application of our strict quantitative selection criteria. Sites with directions most deviant from the mean also had the largest α_{95} values (see Table 2), suggesting that these deviant directions have poor within-site reproducibility and hence low reliability. The mean of the normal sites (cross) and the antipode of the mean of the reversed sites (plus sign) are shown in Figure 8c. Also shown are the directions of the IGRF at the site (evaluated for December 1993 and shown as a triangle) and the geocentric axial dipole (GAD) field (shown as a square). These data are Fisher distributed, and statistical summa-

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ries are listed in Table 3. The *Vw* for these data is 3.2, which is less than the critical value of 6.5 determined through simulation (see Figure 8d). Also, the confidence ellipse for the normal data includes the antipode of the mean of the reversed data. These data are therefore likely to be antipodal.

6. Conclusions

[44] The existing database of published paleomagnetic directions from the Canary Islands suggests that the normal and reversed field directions are not antipodal. Furthermore, the mean directions are significantly different from the direction expected from a geocentric axial dipole. Most of these data were not collected for the purposes of time-averaged field modeling; hence many do not meet the more stringent requirements for such data.

[45] We present here new data collected for the purposes of constraining TAF models. Site means are based on strictly defined characteristic directions from a minimum of four separately oriented samples. The presence of complicated "ill-behaved" demagnetization behavior suggests that stepwise demagnetization should be performed on all specimens for inclusion in the site mean. Anything less can mask underlying complications and lead to scatter that is not of geomagnetic origin.

[46] The directions in this new data set that are most deviant from the average direction, some of which could be considered "excursional," are the least reproducible at the level of the site. We suspect that the deviant directions in our data set may not therefore be accurate representations of the geomagnetic field. This observation is the grounds for suggesting that site means in the time-averaged field data base should be based on at least $N = 4$ samples and that the α_{95} criterion be reduced from 20° to 5°.

[47] The resulting data set suggests that the mean normal and reversed directions are antipodal at the 95% level of confidence. Furthermore, these directions are indistinguishable from the direction expected from a geocentric axial dipole.

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Table A1. Incremental Heating Data for LP1009[a]

Argon IHE Data		^{36}Ar(a)	^{37}Ar	^{38}Ar	^{39}Ar(K)	^{40}Ar(a+r)	Age ± 2σ	^{40}Ar(r), %	Fraction, %	K/Ca
96M0335B	1.00 W 0	0.001749	0.000000	−0.000146	0.017416	0.557601	2.04 ± 2.12	7.32	0.23	no calcium
96M0335C	2.00 W 0	0.004543	0.323999	0.000018	0.198459	1.637503	1.29 ± 0.19	18.03	2.63	0.300
96M0335E	3.00 W 1	0.001650	0.224289	−0.000025	0.155063	0.677436	1.07 ± 0.22	28.02	2.05	0.339
96M0335F	4.00 W 1	0.000733	0.253438	−0.000083	0.196795	0.472113	1.13 ± 0.16	54.12	2.60	0.380
96M0335H	5.00 W 1	0.006457	3.593994	−0.001237	3.188659	5.484742	0.98 ± 0.02	65.18	42.21	0.435
96M0335I	6.00 W 1	0.004657	2.354784	−0.000165	1.833490	3.395924	0.96 ± 0.03	59.45	24.27	0.382
96M0335L	7.00 W 1	0.021560	9.990196	0.001222	1.499766	8.035260	0.97 ± 0.07	20.71	19.85	0.074
96M0335M	8.00 W 1	0.037482	18.753280	0.001403	0.465005	11.791798	1.34 ± 0.39	6.07	6.16	0.012
Total		0.078829	35.493980	0.000987	7.554653	32.052377				

Summary Results	40(a+r)/ 36(a) ± 2σ	40*/39(K) ± 2σ	Age	2σ
Age plateau[b]		1.1166 ± 0.0446 (3.99)	0.971	0.038 (3.993)
Total fusion		1.1593 ± 0.0415 (3.58)	1.008	0.036 (3.580)
Normal isochron[c]	298.7 ± 4.5 (1.49)	1.1058 ± 0.0229 (2.07)	0.962	0.020 (2.068)
Inverse isochron[d]	298.8 ± 4.5 (1.49)	1.1067 ± 0.0227 (2.05)	0.963	0.019 (2.053)

[a]Values in parentheses are the cumulative fraction in percent.
[b]Average K/Ca is 0.27. The cumulative fraction is 97.14%.
[c]Sum is 6.03. Mean weighted standard deviation is 1.51.
[d]Sum is 7.07. Mean weighted standard deviation is 1.77.

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Table A2. Incremental Heating Data for LP1010[a]

Argon IHE Data			^{36}Ar(a)	^{37}Ar(Ca)	^{38}Ar(Cl)	^{39}Ar(K)	^{40}Ar(a + r)	Age ± 2σ	^{40}Ar(r), %	Fraction, %	K/Ca
96M0337B	1.00 W	0	0.004540	0.068644	0.000160	0.044597	1.440973	1.91 ± 1.10	6.90	0.72	0.318
96M0337C	2.00 W	1	0.004590	0.285029	0.000000	0.194791	1.584168	0.99 ± 0.16	14.38	3.14	0.335
96M0337E	3.00 W	1	0.002600	0.453096	0.000000	0.328172	1.124194	0.92 ± 0.10	31.65	5.28	0.355
96M0337F	4.00 W	1	0.005160	1.879713	0.000000	1.721064	3.324057	0.89 ± 0.04	54.10	27.71	0.449
96M0337H	5.00 W	1	0.007020	1.856250	0.000000	2.224497	4.335716	0.86 ± 0.03	52.13	35.82	0.587
96M0337I	6.00 W	1	0.010614	1.088702	0.000071	0.926096	4.057856	0.84 ± 0.06	22.70	14.91	0.417
96M0337L	7.00 W	1	0.020850	3.546651	0.000605	0.419949	6.587416	0.86 ± 0.17	6.47	6.76	0.058
96M0337M	8.00 W	0	0.090813	16.482174	0.001700	0.351261	27.949912	2.69 ± 0.52	3.99	5.66	0.010
Total			0.146186	25.660259	0.002536	6.210428	50.404293				

Summary Results	40(a+r)/ 36(a) ± 2σ	40*/39(K) ± 2σ	Age	2σ
Age plateau[b]		1.0275 ± 0.0243 (2.36)	0.87	0.02 (2.44)
Total fusion		1.1604 ± 0.1011 (8.71)	0.98	0.09 (8.73)
Normal isochron	296.1 ± 4.1 (1.40)	1.0237 ± 0.0343 (3.35)	0.87	0.03 (3.40)
Inverse isochron	296.2 ± 4.2 (1.42)	1.0243 ± 0.0346 (3.38)	0.87	0.03 (3.43)

[a]Values in parentheses are the cumulative fraction in percent.
[b]The cumulative fraction is 93.63%.

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Table A3. Incremental Heating Data for LP1026[a]

Argon IHE Data			^{36}Ar(a)	^{37}Ar(Ca)	^{38}Ar(Cl)	^{39}Ar(K)	^{40}Ar(a + r)	Age ± 2σ	^{40}Ar(r), %	Fraction, %	K/Ca
96M0338A	1.00 W	0	0.001040	0.019078	0.000000	0.007270	0.300859	0.00 ± 2.92	−2.15	0.09	0.187
96M0338B	2.00 W	0	0.002550	0.082370	0.000053	0.034439	0.790136	0.85 ± 0.87	4.63	0.43	0.205
96M0338D	3.00 W	1	0.004180	0.542924	0.000000	0.326435	1.659945	1.01 ± 0.12	25.58	4.07	0.295
96M0338E	4.00 W	1	0.003280	1.007750	0.000000	0.714586	1.828296	0.93 ± 0.06	46.97	8.90	0.347
96M0338G	5.00 W	1	0.004830	2.350163	0.000000	2.015909	3.682841	0.87 ± 0.03	61.22	25.12	0.420
96M0338H	6.00 W	1	0.005570	3.121341	0.000000	2.774842	4.743408	0.87 ± 0.02	65.27	34.58	0.436
96M0338I	7.00 W	1	0.010774	4.814004	0.000229	1.914544	5.270019	0.85 ± 0.02	39.58	23.86	0.195
96M0338L	8.00 W	0	0.008780	4.454810	0.000411	0.237025	3.038193	1.45 ± 0.10	14.60	2.95	0.026
Total			0.041004	16.392439	0.000693	8.025050	21.313697				

Summary Results	40(a+r)/ 36(a) ± 2 σ	40*/39(K) ± 2σ	Age	2σ
Age plateau[b]		1.1130 ± 0.0294 (2.64)	0.86	0.02 (2.71)
Total fusion		1.1460 ± 0.0483 (4.21)	0.89	0.04 (4.26)
Normal isochron	295.7 ± 17.3 (5.85)	1.1109 ± 0.0700 (6.30)	0.86	0.05 (6.33)
Inverse isochron	296.8 ± 17.8 (6.01)	1.1084 ± 0.0716 (6.46)	0.86	0.06 (6.48)

[a]Values in parentheses are the cumulative fraction in percent.
[b]The cumulative fraction is 96.53%.

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Table A4. Incremental Heating Data for LP1030[a]

Argon IHE Data			^{36}Ar(a)	^{37}Ar(Ca)	^{38}Ar(Cl)	^{39}Ar(K)	^{40}Ar(a + r)	Age ± 2σ	^{40}Ar(r), %	Fraction, %	K/Ca
96M0339A	1.00 W	0	0.002270	0.183562	0.000076	0.096667	0.836334	1.28 ± 0.45	19.79	2.05	0.258
96M0339C	2.00 W	0	0.002200	1.025588	0.000000	0.755674	1.793021	1.13 ± 0.05	63.72	16.02	0.361
96M0339D	3.00 W	1	0.000924	1.421724	0.000000	1.189335	1.850277	1.00 ± 0.01	85.20	25.21	0.410
96M0339F	4.00 W	1	0.000873	1.586966	0.000000	1.346083	2.093341	1.02 ± 0.02	87.63	28.54	0.416
96M0339G	5.00 W	1	0.000919	1.061238	0.000043	0.794481	1.375654	1.04 ± 0.03	80.22	16.84	0.367
96M0339I	6.00 W	0	0.000799	1.617147	0.000425	0.318811	0.756286	1.22 ± 0.06	68.76	6.76	0.097
96M0339J	7.00 W	0	0.002050	4.825018	0.000805	0.215792	0.841421	0.82 ± 0.17	28.00	4.57	0.022
Total			0.010035	11.721243	0.001349	4.716843	9.546334				

Summary Results	40(a+r)/ 36(a) ± 2σ	40*/39(K) ± 2σ	Age	2σ
Age plateau[b]		1.3425 ± 0.0334 (2.49)	1.01	0.03 (2.56)
Total fusion		1.3952 ± 0.0578 (4.14)	1.05	0.04 (4.18)
Normal isochron	384.3 ± 60.9 (15.84)	1.2661 ± 0.0594 (4.69)	0.95	0.04 (4.73)
Inverse isochron	382.9 ± 71.4 (18.66)	1.2717 ± 0.0674 (5.30)	0.95	0.05 (5.33)

[a]Values in parentheses are the cumulative fraction in percent.
[b]The cumulative fraction is 70.60%.

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